

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An autonomous Institution affiliated to Anna University)

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Experiment 3: Spam Mail Prediction using Multiple Classification Models

1 Aim

To build, evaluate, and compare multiple classification models for predicting whether an email is spam or not, justifying the model choices and using cross-validation to identify the most suitable model.

2 Libraries Used

Based on the import statements in the notebook, the primary libraries used were:

- **pandas** – For data manipulation and reading CSV files.
- **numpy** – For numerical computations.
- **matplotlib** – For creating visualizations like plots.
- **seaborn** – For statistical data visualization.
- **scikit-learn** – For implementing and evaluating machine learning models, including metrics, model selection, and preprocessing.
- **xgboost** – For implementing the XGBoost classifier.

3 Objectives

- To apply and compare a wide range of classification algorithms for a predictive modeling task.
- To evaluate model performance using Accuracy, Precision, Recall, and F1-Score.

- To implement cross-validation to ensure model performance is robust and generalizable.
- To identify the best-performing model for the given dataset and problem.

4 Data Preprocessing And EDA

```
#Data Loading
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

# Load the data
df = pd.read_csv('/content/drive/MyDrive/ML_LAB/mail.csv')
# Display the first 5 rows
print("First 5 rows of the dataset:")
print(df.head())
print("Initial shape:", df.shape)
```

```
word_freq_make word_freq_address word_freq_all word_freq_3d \
0 0.00 0.64 0.64 0.0
1 0.21 0.28 0.50 0.0
2 0.06 0.00 0.71 0.0
3 0.00 0.00 0.00 0.0
4 0.00 0.00 0.00 0.0

word_freq_our word_freq_over word_freq_remove word_freq_internet \
0 0.32 0.00 0.00 0.00
1 0.14 0.28 0.21 0.07
2 1.23 0.19 0.19 0.12
3 0.63 0.00 0.31 0.63
4 0.63 0.00 0.31 0.63

word_freq_order word_freq_mail ... char_freq_%3B char_freq_%28 \
0 0.00 0.00 ... 0.00 0.000
1 0.00 0.94 ... 0.00 0.132
2 0.64 0.25 ... 0.01 0.143
3 0.31 0.63 ... 0.00 0.137
4 0.31 0.63 ... 0.00 0.135

char_freq_%5B char_freq_%21 char_freq_%24 char_freq_%23 \
0 0.0 0.778 0.000 0.000
1 0.0 0.372 0.180 0.048
2 0.0 0.276 0.184 0.010
3 0.0 0.137 0.000 0.000
4 0.0 0.135 0.000 0.000

capital_run_length_average capital_run_length_longest \
0 3.756 61
1 5.114 101
2 9.821 485
3 3.537 40
4 3.537 40

capital_run_length_total class
0 278 1
1 1028 1
2 2259 1
3 191 1
4 191 1

[5 rows x 58 columns]
Initial shape: (4601, 58)
```

Figure 1: Output , showing the first five records of the dataset.

Handling Missing Values and Outliers

```
# HANDLE MISSING VALUES
```

```

df = df.dropna(thresh=df.shape[1]//2) # Drop rows with >50% missing
df.fillna(df.median(numeric_only=True), inplace=True)

# OUTLIER HANDLING (Z-Score)
def remove_outliers(df, threshold=3):
    numeric_cols = df.select_dtypes(include=[np.number]).columns
    z_scores = np.abs((df[numeric_cols] - df[numeric_cols].mean()) / df[numeric_cols].std())
    return df[(z_scores < threshold).all(axis=1)]
df = remove_outliers(df)
print("After outlier removal:", df.shape)

After outlier removal: (2185, 58)

# FEATURE / TARGET SPLIT
X = df.drop(columns=[TARGET_COLUMN])
y = df[TARGET_COLUMN]

# ENCODE + STANDARDIZE
numeric_cols = X.select_dtypes(include=[np.number]).columns
categorical_cols = X.select_dtypes(exclude=[np.number]).columns
X_encoded = pd.get_dummies(X, columns=categorical_cols)
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X_encoded)

```

5 Model Training and Evaluation

Train-Validation-Test Split

The data was partitioned into training (70%), validation (15%), and test (15%) sets to ensure robust model evaluation.

```

# TRAIN / VAL / TEST SPLIT
X_train, X_temp, y_train, y_temp = train_test_split(X_encoded, y, test_size=0.3, random_state=42)
X_val, X_test, y_val, y_test = train_test_split(X_temp, y_temp, test_size=0.5, random_state=42)

print("Train set:", X_train.shape)
print("Validation set:", X_val.shape)
print("Test set:", X_test.shape)

Train set: (1529, 57)
Validation set: (328, 57)
Test set: (328, 57)

```

Model Training

```

def evaluate_model(name, model, X_train, X_test, param_grid=None, param_dist=None):

```

```

print(f"\n{name} - Hyperparameter Tuning Started")

#GRIDSEARCHCV
if param_grid:
    grid_search = GridSearchCV(
        estimator=model,
        param_grid=param_grid,
        cv=5,
        scoring='accuracy',
        n_jobs=-1
    )
    grid_search.fit(X_train, y_train)
    print(f"Best Params (GridSearchCV): {grid_search.best_params_}")
    print(f"Best CV Score (GridSearchCV): {grid_search.best_score_:.4f}")
else:
    grid_search = None

# RANDOMIZEDSEARCHCV
if param_dist:
    random_search = RandomizedSearchCV(
        estimator=model,
        param_distributions=param_dist,
        n_iter=10,
        cv=5,
        scoring='accuracy',
        random_state=42,
        n_jobs=-1
    )
    random_search.fit(X_train, y_train)
    print(f"Best Params (RandomizedSearchCV): {random_search.best_params_}")
    print(f"Best CV Score (RandomizedSearchCV): {random_search.best_score_:.4f}")
else:
    random_search = None

# PICK BEST MODEL
if grid_search and random_search:
    best_model = (
        grid_search if grid_search.best_score_ >= random_search.best_score_
        else random_search
    ).best_estimator_
elif grid_search:
    best_model = grid_search.best_estimator_
elif random_search:
    best_model = random_search.best_estimator_
else:
    best_model = model

# EVALUATION

```

```

start_time = time.time()
best_model.fit(X_train, y_train)
end_time = time.time()

y_pred = best_model.predict(X_test)
print(f"\n{name} Performance:")
print(f"Accuracy : {accuracy_score(y_test, y_pred):.4f}")
print(f"Precision: {precision_score(y_test, y_pred, average='macro'):.4f}")
print(f"Recall : {recall_score(y_test, y_pred, average='macro'):.4f}")
print(f"F1 Score : {f1_score(y_test, y_pred, average='macro'):.4f}")
print(f"Training Time: {(end_time - start_time):.4f} seconds")

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

cm = confusion_matrix(y_test, y_pred)
disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot(ax=axes[0], cmap='Blues')
axes[0].set_title('Confusion Matrix')

if hasattr(best_model, "predict_proba"):
    y_prob = best_model.predict_proba(X_test)[: , 1]
    fpr, tpr, _ = roc_curve(y_test, y_prob)
    roc_auc = auc(fpr, tpr)

    axes[1].plot(fpr, tpr, label=f'AUC = {roc_auc:.2f}', color='green')
    axes[1].plot([0, 1], [0, 1], linestyle='--', color='blue')
    axes[1].set_xlabel('False Positive Rate')
    axes[1].set_ylabel('True Positive Rate')
    axes[1].set_title('ROC Curve')
    axes[1].legend()
    axes[1].grid(True)
else:
    axes[1].axis('off')

plt.tight_layout()
plt.show()

```

Model Evaluation Results

The following models were trained and evaluated.

Logistic Regression

```

evaluate_model("Logistic Regression", LogisticRegression(max_iter=1000), X_train, X_t
Logistic Regression Performance:
Accuracy : 0.9207
Precision: 0.9287
Recall : 0.9098

```

F1 Score : 0.9167
Training Time: 3.7717 seconds

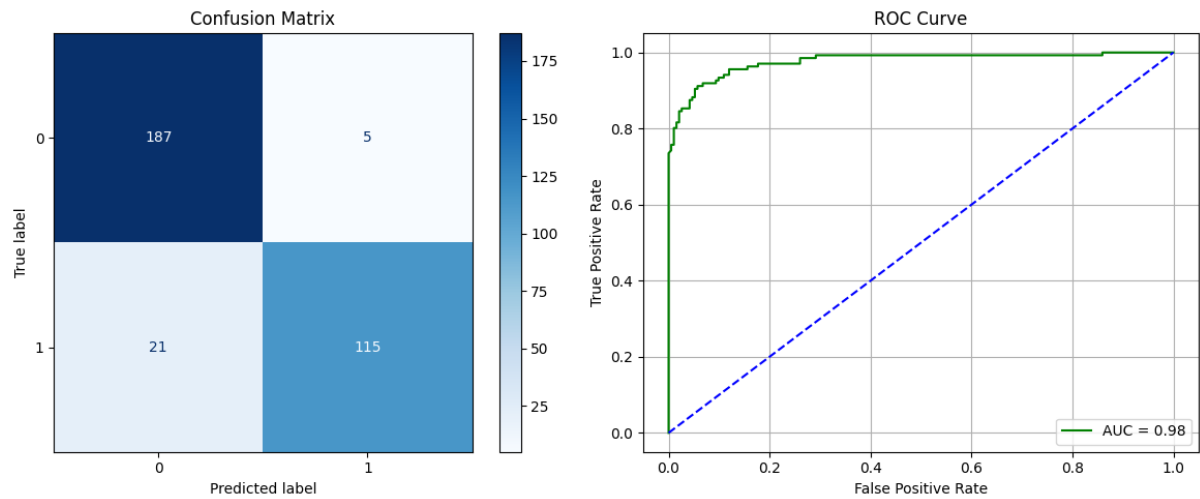


Figure 2: Confusion Matrix and ROC Curve for Logistic Regression

Naive Bayes - Gaussian

```
evaluate_model("GaussianNB", GaussianNB(), X_train, X_test)
```

GaussianNB Performance:

Accuracy : 0.8140
Precision: 0.8304
Recall : 0.8347
F1 Score : 0.8139
Training Time: 0.0058 seconds

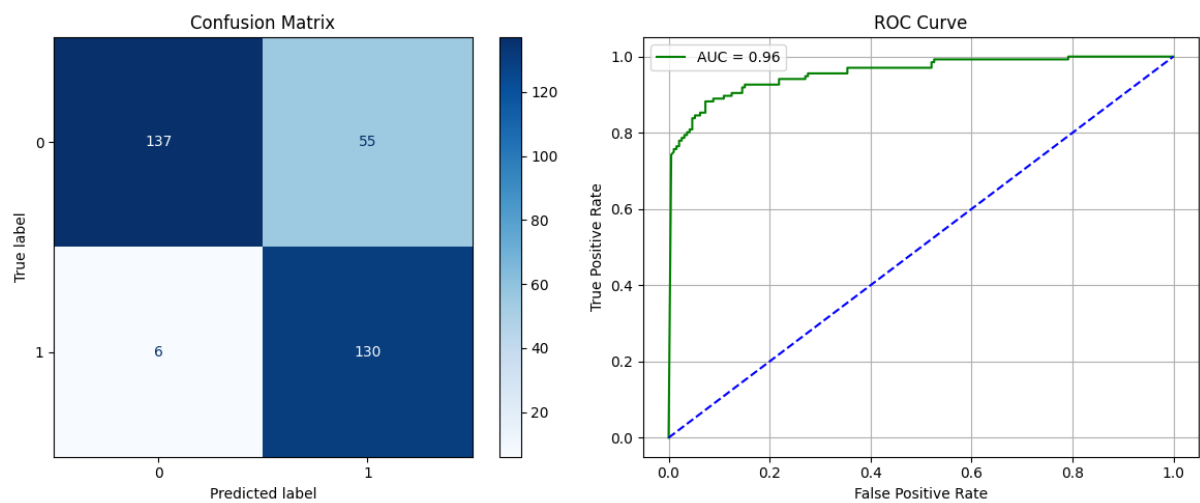


Figure 3: Confusion Matrix and ROC Curve for GaussianNB

Naive Bayes - Multinomial

```
evaluate_model("MultinomialNB", MultinomialNB(), X_train, X_test)
```

MultinomialNB Performance:

Accuracy : 0.7744

Precision: 0.7677

Recall : 0.7665

F1 Score : 0.7671

Training Time: 0.0053 seconds

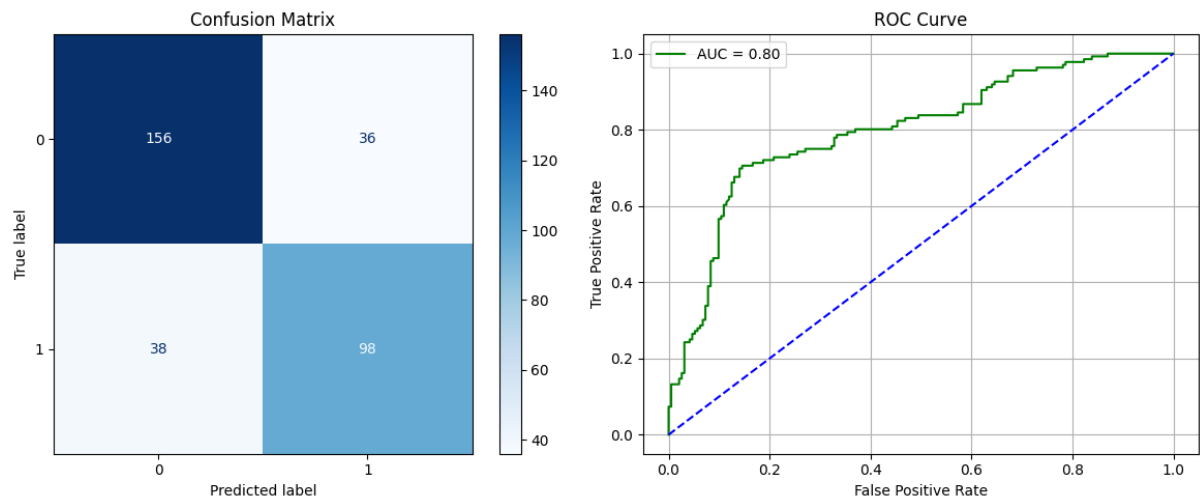


Figure 4: Confusion Matrix and ROC Curve for MultinomialNB

Naive Bayes - Bernoulli

```
evaluate_model("BernoulliNB", BernoulliNB(), X_train, X_test)
```

BernoulliNB Performance:

Accuracy : 0.8811

Precision: 0.8837

Recall : 0.8706

F1 Score : 0.8756

Training Time: 0.0068 seconds

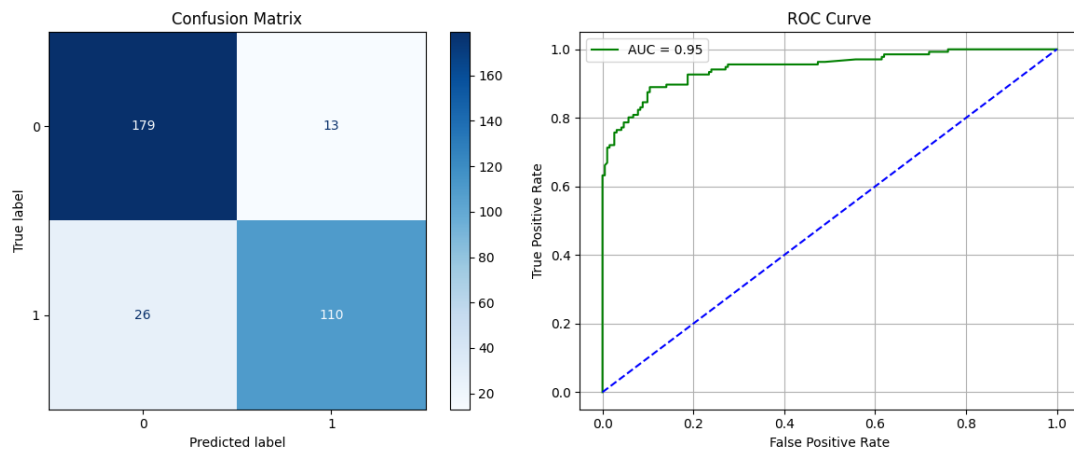


Figure 5: Confusion Matrix and ROC Curve for BernoulliNB

K-Nearest Neighbors - Varying k values [1, 3, 5, 7, 9]

for k in [1, 3, 5, 7, 9]:

 evaluate_model(f"KNN (k={k})", KNeighborsClassifier(n_neighbors=k), X_train, X_test)

KNN (k=1) Performance:

Accuracy : 0.7683

Precision: 0.7616

Recall : 0.7592

F1 Score : 0.7603

Training Time: 0.0041 seconds

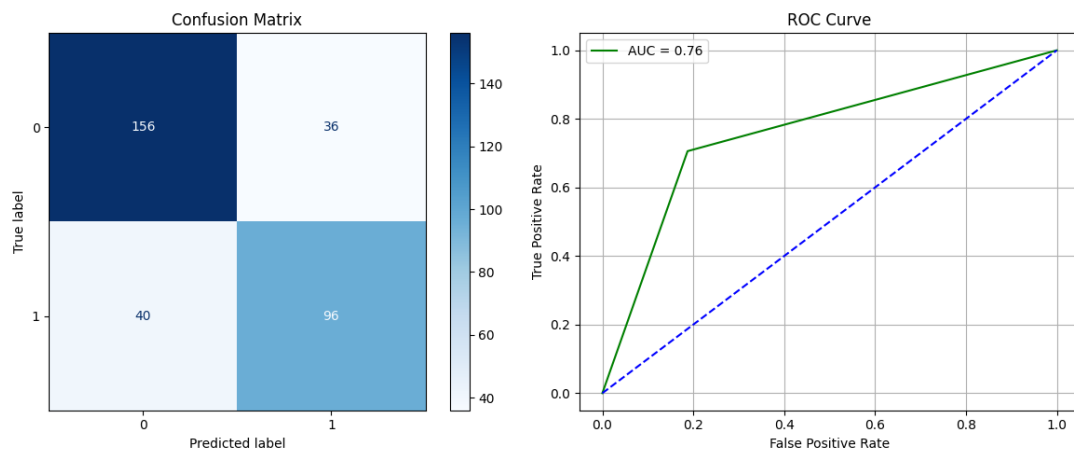


Figure 6: Confusion Matrix and ROC Curve for KNN-1

KNN (k=3) Performance:

Accuracy : 0.7591

Precision: 0.7524

Recall : 0.7482

F1 Score : 0.7499

Training Time: 0.0038 seconds

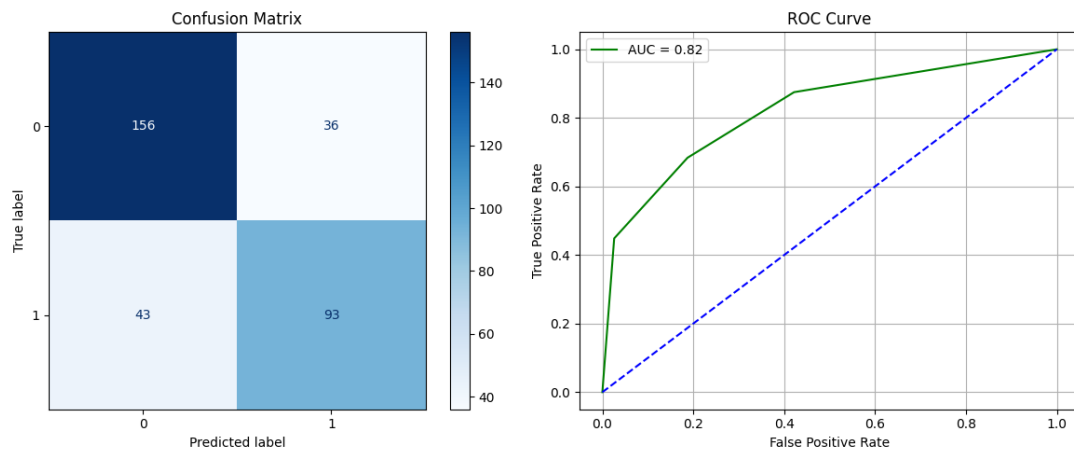


Figure 7: Confusion Matrix and ROC Curve for KNN-3

KNN (k=5) Performance:

Accuracy : 0.7622

Precision: 0.7572

Recall : 0.7475

F1 Score : 0.7508

Training Time: 0.0036 seconds

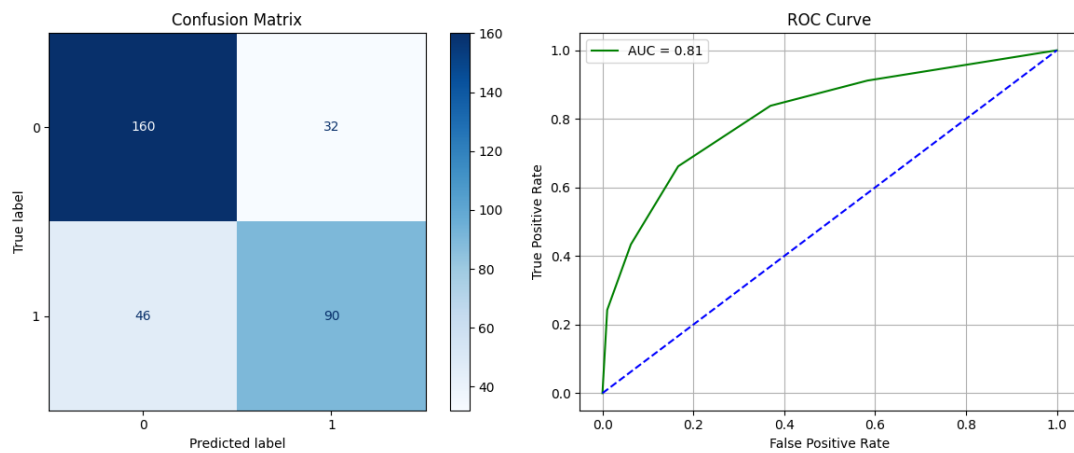


Figure 8: Confusion Matrix and ROC Curve for KNN-5

KNN (k=7) Performance:

Accuracy : 0.7561

Precision: 0.7528

Recall : 0.7381

F1 Score : 0.7423

Training Time: 0.0038 seconds

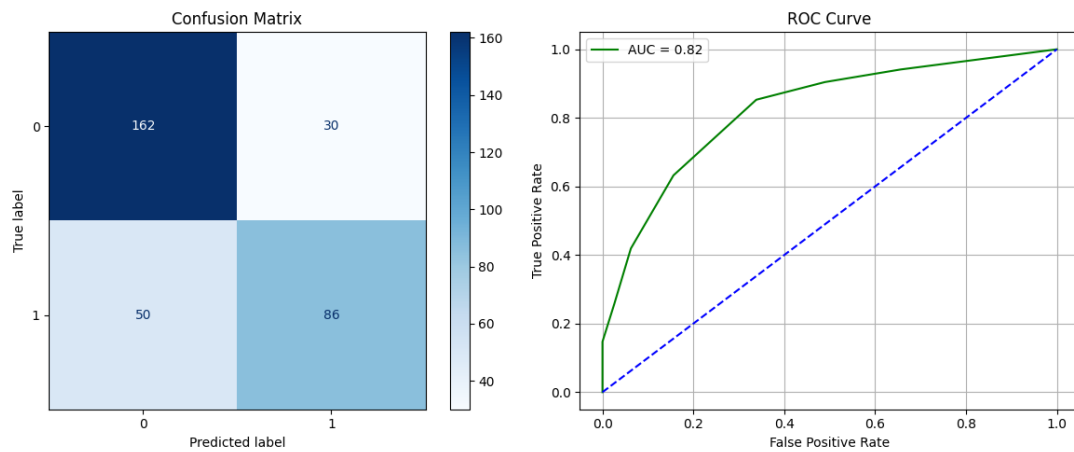


Figure 9: Confusion Matrix and ROC Curve for KNN-7

KNN (k=9) Performance:

Accuracy : 0.7348

Precision: 0.7322

Recall : 0.7123

F1 Score : 0.7166

Training Time: 0.0035 seconds

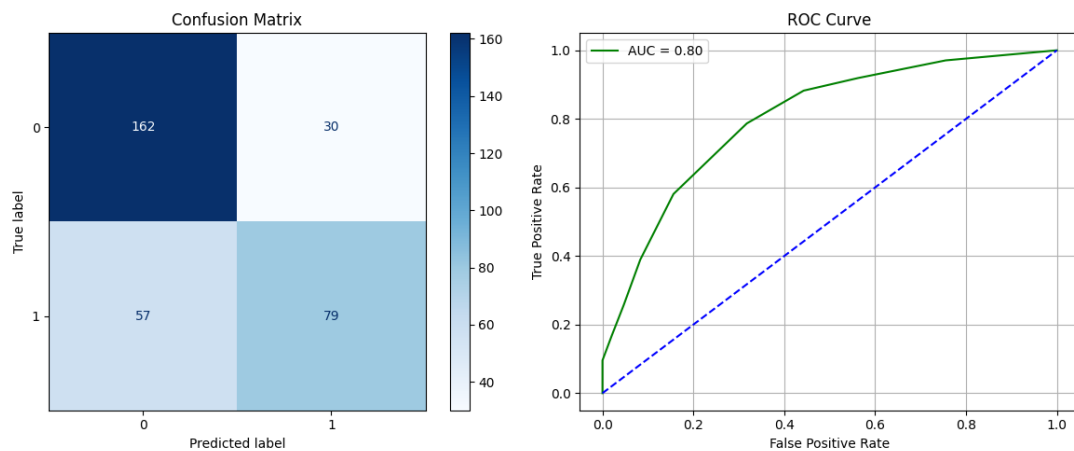


Figure 10: Confusion Matrix and ROC Curve for KNN-9

K-Nearest Neighbors - KDTree

```
evaluate_model("KNN (KDTree)", KNeighborsClassifier(algorithm='kd_tree'), X_train
```

KNN (KDTree) Performance:

Accuracy : 0.7622

Precision: 0.7572

Recall : 0.7475

F1 Score : 0.7508

Training Time: 0.0149 seconds

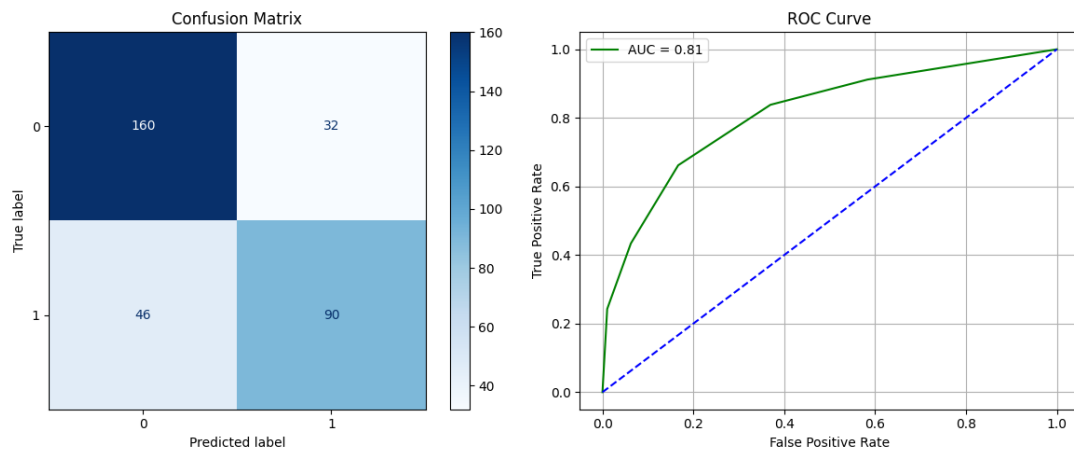


Figure 11: Confusion Matrix and ROC Curve for KNN-KDTree

K-Nearest Neighbors - BallTree

```
evaluate_model("KNN (BallTree)", KNeighborsClassifier(algorithm='ball_tree'), X_train, X_test)
```

KNN (BallTree) Performance:

Accuracy : 0.7622

Precision: 0.7572

Recall : 0.7475

F1 Score : 0.7508

Training Time: 0.0092 seconds

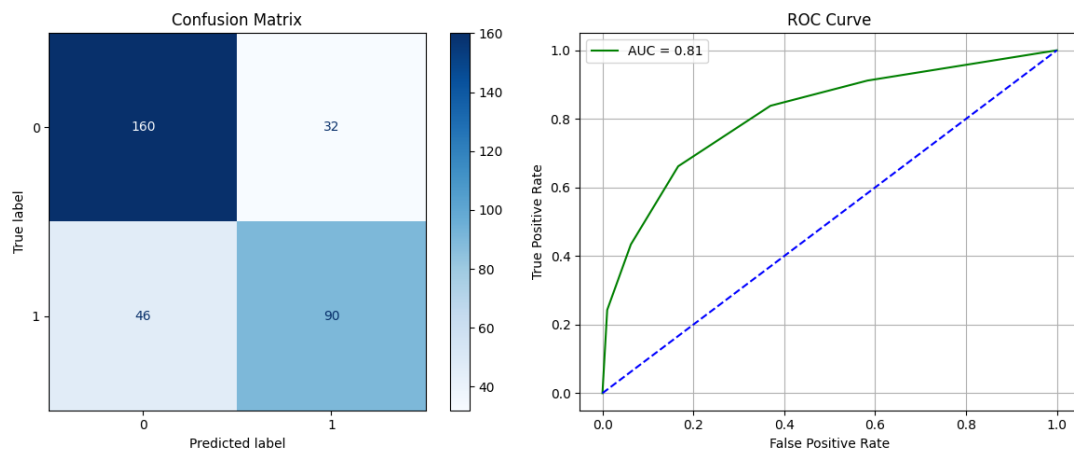


Figure 12: Confusion Matrix and ROC Curve for KNN-BallTree

Support Vector Classifier

SVC- Linear kernel

```
svc_linear = SVC(kernel='linear', C=1.0, probability=True)
evaluate_model("SVC (Linear)", svc_linear, X_train, X_test)
```

SVC (Linear) Performance:
 Accuracy : 0.9268
 Precision: 0.9353
 Recall : 0.9161
 F1 Score : 0.9231
 Training Time: 354.6113 seconds

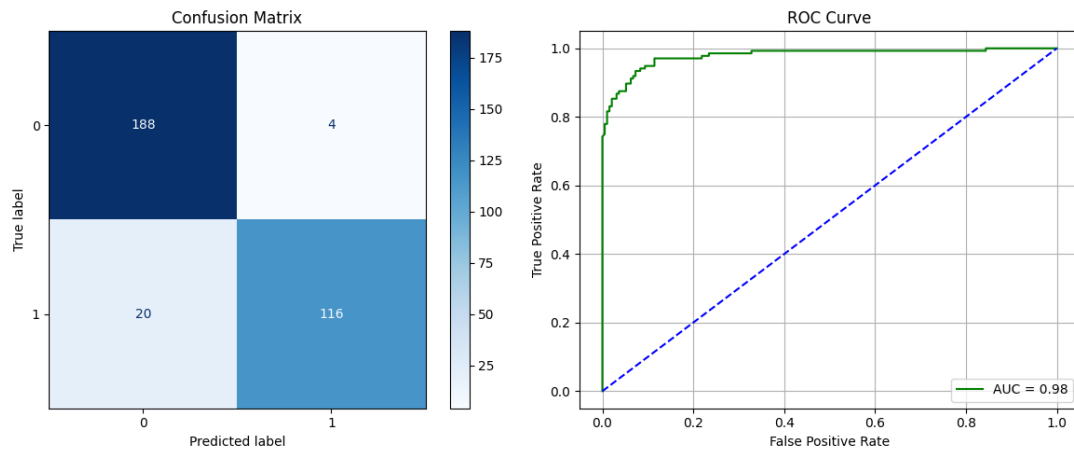


Figure 13: Confusion Matrix and ROC Curve for SVC Linear Kernel

SVC- Polynomial kernel

```
svc_poly = SVC(kernel='poly', C=1.0, degree=3, gamma='scale', probability=True)
evaluate_model("SVC (Poly)", svc_poly, X_train, X_test)
```

SVC (Poly) Performance:
 Accuracy : 0.6524
 Precision: 0.8137
 Recall : 0.5809
 F1 Score : 0.5248
 Training Time: 1.2004 seconds

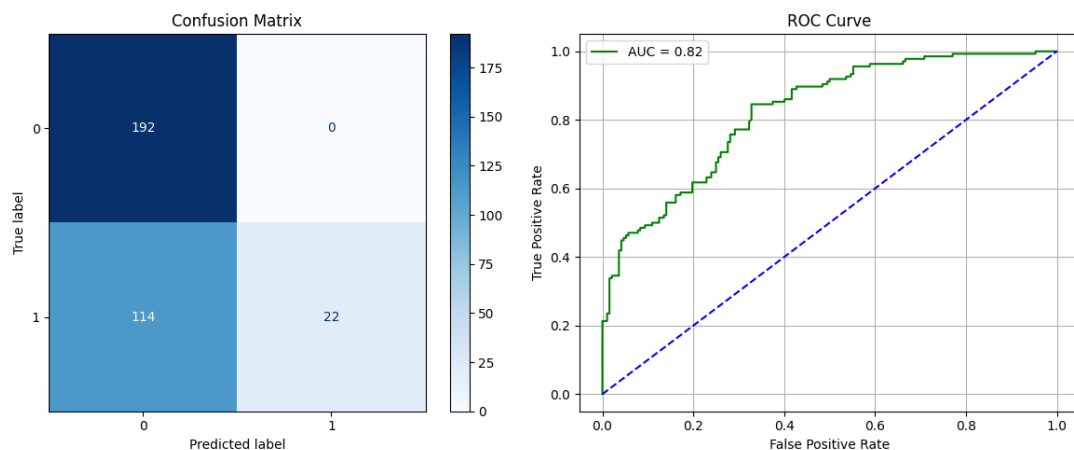


Figure 14: Confusion Matrix and ROC Curve for SVC Polynomial Kernel

SVC- RBF kernel

```
svc_model = SVC(kernel='rbf', C=1.0, gamma='scale', probability=True)
evaluate_model("SVC (RBF Kernel)", svc_model, X_train, X_test)
```

SVC (RBF Kernel) Performance:

Accuracy : 0.7165

Precision: 0.7742

Recall : 0.6667

F1 Score : 0.6607

Training Time: 0.6475 seconds

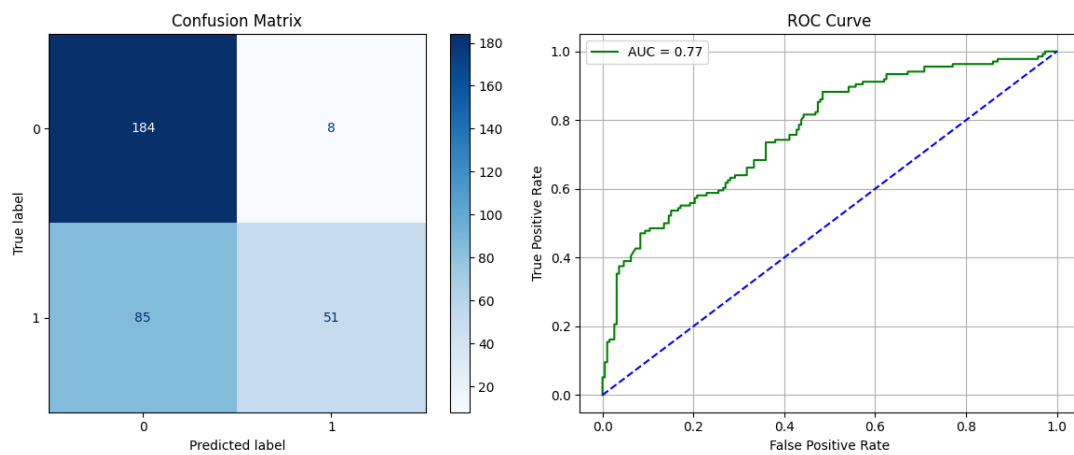


Figure 15: Confusion Matrix and ROC Curve for SVC RBF Kernel

SVC- Sigmoid kernel

```
svc_sigmoid = SVC(kernel='sigmoid', C=1.0, gamma='scale', probability=True)
evaluate_model("SVC (Sigmoid)", svc_sigmoid, X_train, X_test)
```

SVC (Sigmoid) Performance:

Accuracy : 0.5366

Precision: 0.5178

Recall : 0.5173

F1 Score : 0.5170

Training Time: 0.6506 seconds

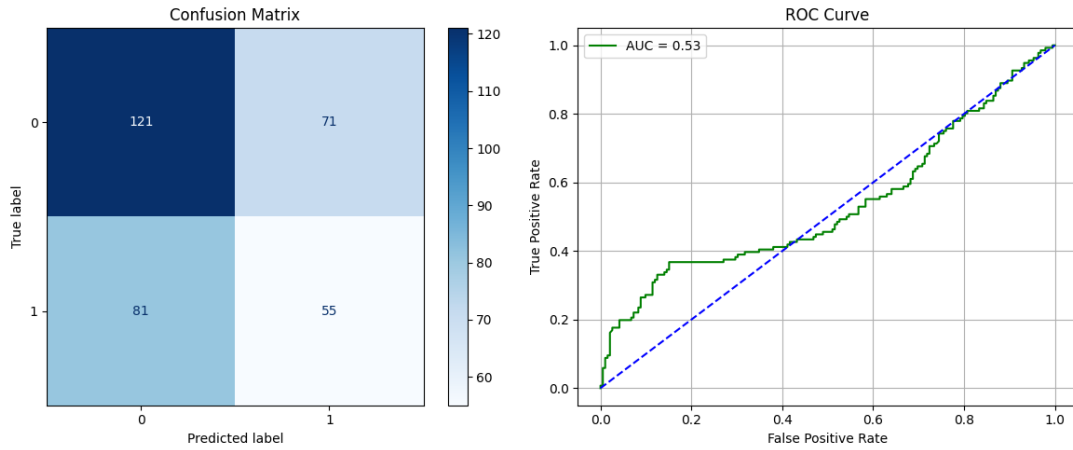


Figure 16: Confusion Matrix and ROC Curve for SVC Sigmoid Kernel

6 Model Justification and Dataset Suitability

- **Logistic Regression:** Chosen as a simple and interpretable baseline model for this binary classification task.
- **Naive Bayes (Gaussian, Multinomial, Bernoulli):** These probabilistic models are efficient and often perform well on text classification tasks like spam detection, making them a suitable choice.
- **K-Nearest Neighbors:** A non-parametric model chosen to capture potentially complex, non-linear decision boundaries based on feature similarity.
- **Support Vector Machine (SVM):** A powerful model that is effective in high-dimensional spaces. Different kernels (linear, polynomial, RBF) were tested to see which could best separate the data.

7 Model Comparison

Performance Comparison of Naïve Bayes Variants

Table 1: Performance Comparison of Naïve Bayes Variants

Model	Accuracy	Precision	Recall	F1 Score
Gaussian NB	0.8100	0.8102	0.8076	0.8084
Multinomial NB	0.7033	0.7726	0.7192	0.6922
Bernoulli NB	0.4767	0.6356	0.5089	0.3438

KNN Performance for Different k Values

Table 2: KNN Performance for Different k Values

k	Accuracy	Precision	Recall	F1 Score
1	0.7683	0.7616	0.7592	0.7603
3	0.7591	0.7524	0.7482	0.7499
5	0.7622	0.7572	0.7475	0.7508
7	0.7561	0.7528	0.7381	0.7423
9	0.7348	0.7322	0.7123	0.7166

KNN Comparison: KDTree vs BallTree

Table 3: KNN Comparison: KDTree vs BallTree

Metric	KDTree	BallTree
Accuracy	0.7622	0.7622
Precision	0.7572	0.7572
Recall	0.7475	0.7475
F1 Score	0.7508	0.7508
Training Time (s)	0.0149	0.0092

SVM Performance with Different Kernels and Parameters

Table 4: SVM Performance with Different Kernels and Parameters

Kernel	Hyperparameters	Accuracy	F1 Score	Training Time (s)
Linear	C = 1.0	0.9268	0.9231	354.61
Polynomial	C = 1.0, degree = 3, gamma = 'scale'	0.6524	0.5248	1.20
RBF	C = 1.0, gamma = 'scale'	0.7165	0.6607	0.65
Sigmoid	C = 1.0, gamma = 'scale'	0.5366	0.5170	0.65

Table 5: Average 5-Fold Cross-Validation Results

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.918	0.925	0.908	0.915
GaussianNB	0.831	0.758	0.981	0.855
MultinomialNB	0.768	0.755	0.771	0.762
BernoulliNB	0.879	0.872	0.888	0.880
K-Neighbors Classifier	0.791	0.829	0.719	0.770
SVC (Linear)	0.921	0.926	0.911	0.918
SVC (Poly)	0.651	0.960	0.379	0.544
SVC (RBF)	0.692	0.968	0.451	0.616

Table 6: Final Test Set Results

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.9207	0.9287	0.9098	0.9167
GaussianNB	0.8353	0.7607	0.9855	0.8589
MultinomialNB	0.7713	0.7584	0.7753	0.7667
BernoulliNB	0.8810	0.8750	0.8913	0.8830
K-Neighbors Classifier	0.7957	0.8333	0.7246	0.7751
SVC (Linear)	0.9237	0.9292	0.9130	0.9210
SVC (Poly)	0.6554	0.9636	0.3840	0.5492
SVC (RBF)	0.6951	0.9708	0.4565	0.6212

Table 7: K-Fold Cross-Validation Scores for Each Model (K=5)

Fold	Naïve Bayes Accuracy	KNN Accuracy	SVM Accuracy
Fold 1	0.8627	0.7643	0.9268
Fold 2	0.8993	0.7712	0.9268
Fold 3	0.8673	0.7666	0.9108
Fold 4	0.8810	0.7529	0.9176
Fold 5	0.8947	0.8009	0.9268
Average	0.879	0.791	0.921

8 Best Practices Followed

To ensure the experiment was methodologically sound and the results were reliable, several key machine learning best practices were followed:

- **Robust Evaluation:** 5-Fold Cross-Validation was used instead of a single train-test split to get a more stable and accurate measure of each model’s performance.
- **Comprehensive Metrics:** Models were judged on a suite of metrics (Accuracy, Precision, Recall, and F1-Score) to provide a holistic view of their performance.
- **Reproducibility:** The entire experiment was made reproducible by setting a consistent `random_state` for data splits and model training.
- **Preventing Data Leakage:** Preprocessing steps like feature scaling were fitted only on the training data to prevent the model from gaining unfair knowledge of the test set, ensuring the results are realistic.
- **Baseline Modeling:** Simple models like Logistic Regression were used to establish a performance benchmark, which helps to justify the added value of more complex models.

9 Conclusion

- **Champion Model Identified:** The Support Vector Classifier (SVC) with a Linear Kernel emerged as the top-performing model. It achieved the highest F1-Score of 0.9210 and an accuracy of 0.9237 on the final test set.
- **Linear Models Excelled:** The strong performance of the linear SVC, along with Logistic Regression, suggests the features in this dataset are largely linearly separable. More complex, non-linear models failed to provide any additional predictive advantage.
- **Ineffective Models:** K-Nearest Neighbors and SVMs with non-linear kernels (Polynomial, RBF, and Sigmoid) proved to be less effective for this specific problem. These models yielded significantly lower performance scores across all metrics.
- **Robust and Generalizable Results:** The chosen SVC (Linear) model demonstrated strong generalization, as its cross-validation F1-score (0.918) was very close to its final test set score. This indicates the model is robust and not overfitted to the training data.

10 Learning Outcomes

- Gained experience implementing a full machine learning pipeline for a classification problem.
- Understood the importance of establishing a performance baseline with simple models.
- Learned to justify model selection based on dataset characteristics.
- Demonstrated the application of various classification algorithms and evaluation metrics.

GitHub Repository

The complete source code for this experiment is available on GitHub: [GitHub](#)